

MDIA-2109
Dynamic Content Design
Week 12: The Document Object
Model

Agenda

What is the Document Object Model?

Adding JS to your HTML

Adding and removing elements

Manipulating elements, styles, and attributes

Event handling



**JavaScript interacts with the
browser through the
Document Object Model**

The DOM

The Document Object Model (DOM) is how JavaScript talks to your webpage

The DOM is a programming interface that represents your HTML as a tree of objects (called nodes) that JavaScript can read and change

It is created by the browser, based on your HTML, and updated live as the page changes



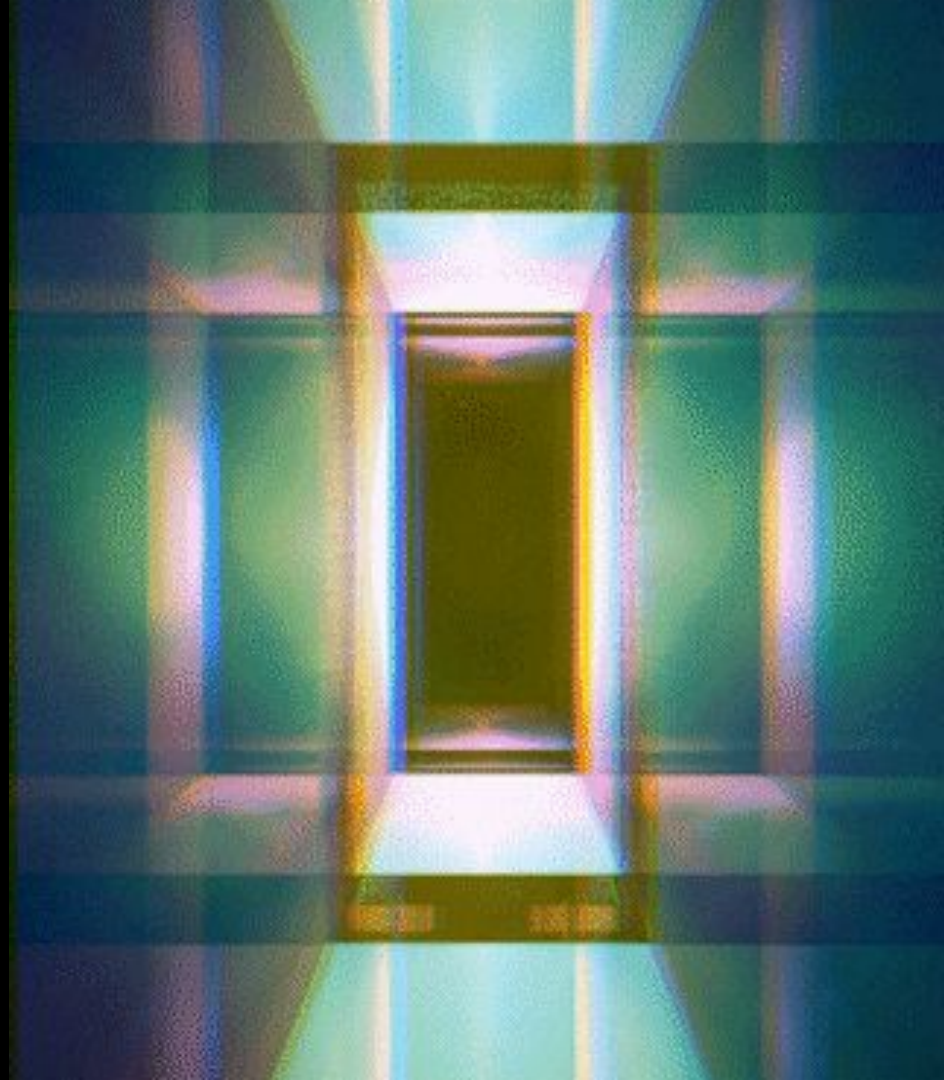
The DOM

The DOM is the bridge between HTML and JS

Without the DOM, JS couldn't:

- Change text
- React to clicks
- Animate things
- Update content dynamically

With the DOM, JS can read, modify, add, and remove nodes



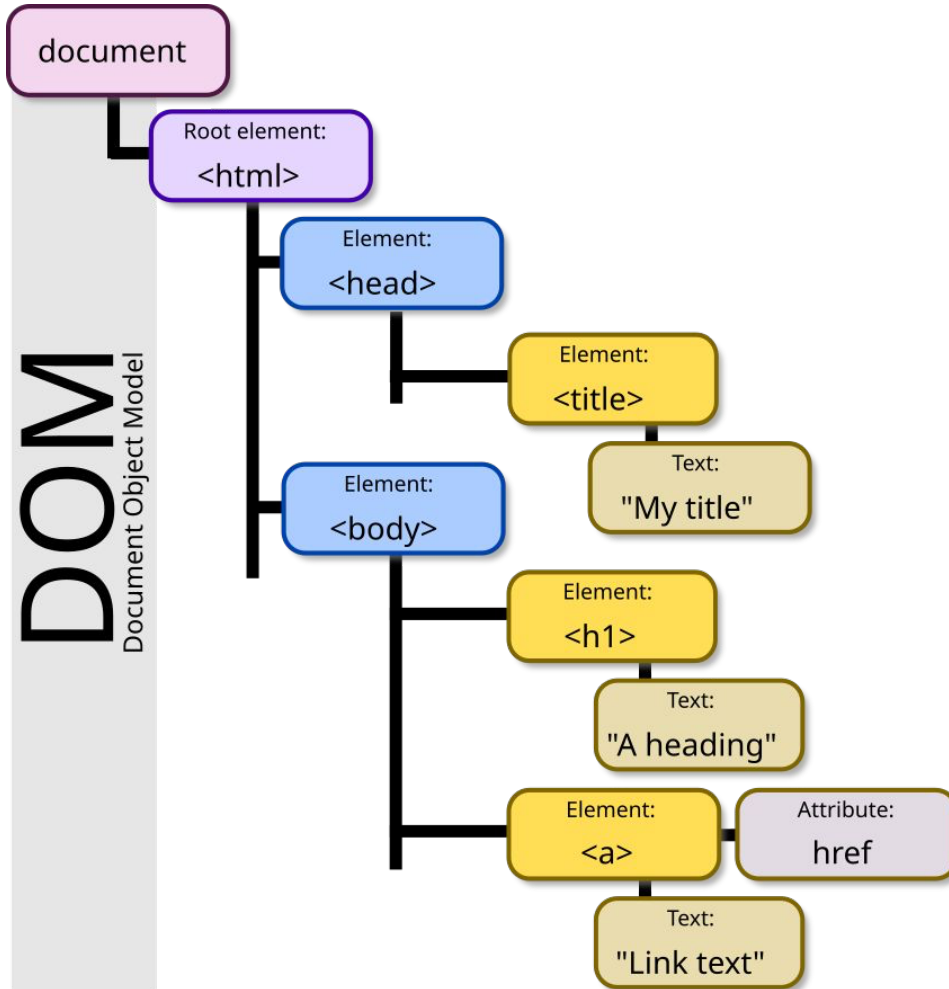
HTML to DOM

```
<html>
  <head>
    <title>A heading</title>
  </head>
  <body>
    <h1>A heading</h1>
    <a href="/">Link text</a>
  </body>
</html>
```

Each node represents part of the HTML document (e.g.: an element, some text, an attribute)

JS interacts with these nodes!





**With the DOM, JS can read,
modify, add, and remove nodes.**

**You can add JS to an HTML document
using the `<script>` tag.**

Adding JS

```
// Inline
<html>
  <body>
    ...
    <script>
      console.log("JS!!!!");
    </script>
  </body>
</html>

// From a file
<html>
  <body>
    ...
    <script src="main.js"></script>
  </body>
</html>
```



Selecting elements

The `document` object is the entry point to the DOM provided by the browser.

It exists globally in any JS running in a browser, and you can use it to select elements (nodes):

```
// Returns first match  
document.querySelector(".link");
```

```
// Returns a NodeList of all matches //  
NodeList is similar to an array  
document.querySelectorAll(".link");
```



Selecting elements

```
let heading = document.querySelector("h1");  
console.log(heading);
```

1. Browser finds the `<h1>`
2. JS gets a reference to it
3. You can now change it

(You don't need to `console.log` to do this, I just wanted you to "see" the element in the browser console)

Creating elements

// Before...

```
<html>
  <body>
    <h1>A heading</h1>
    <a href="/">Link text</a>
  </body>
</html>
```

```
let p = document.createElement("p");
p.textContent = "Hello!";
document.body.appendChild(p);
```

// After...

```
<html>
  <body>
    ...
    <p>Hello!</p>
  </body>
</html>
```


Removing elements

// Before...

```
<html>
  <body>
    <h1>A heading</h1>
    <a href="/">Link text</a>
    <p>Hello!</p>
  </body>
</html>
```

p.remove();

// After...

```
<html>
  <body>
    <h1>A heading</h1>
    <a href="/">Link text</a>
  </body>
</html>
```


Changing elements

// Before...

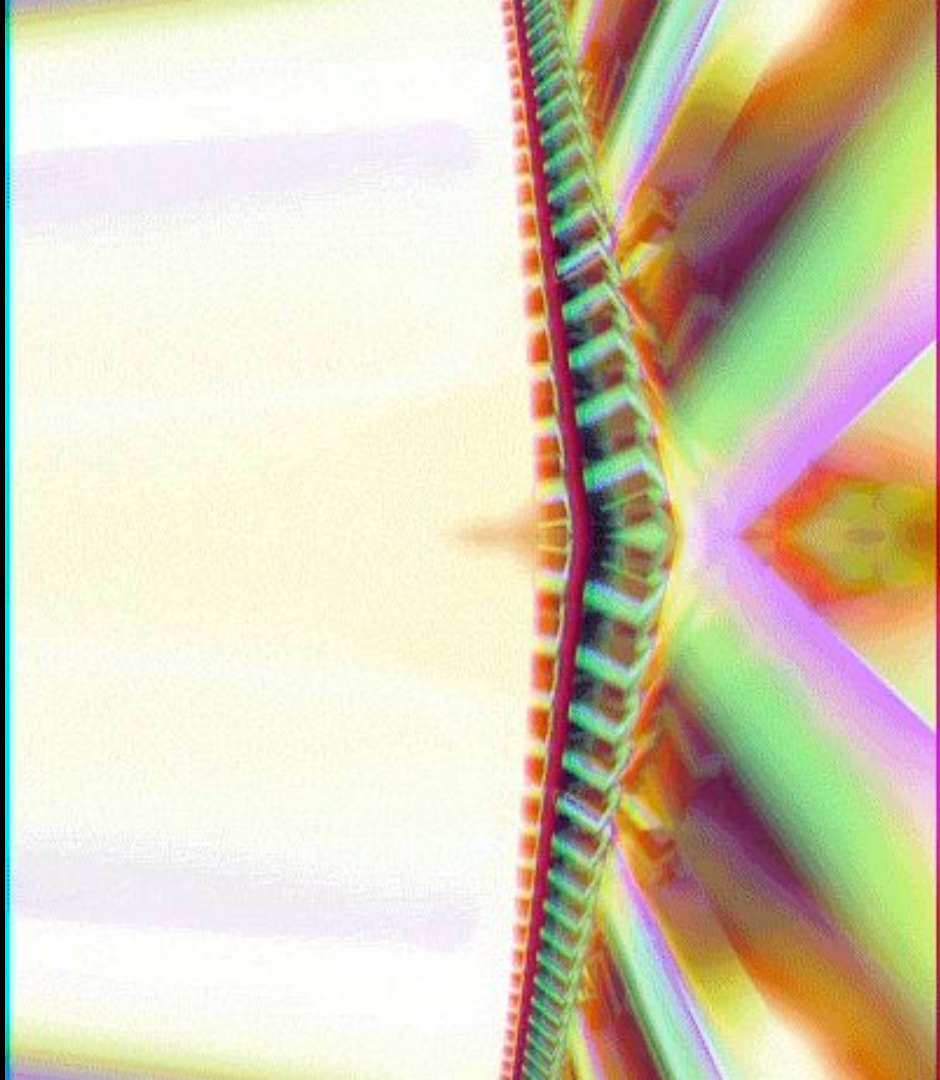
```
<html>
  <body>
    <h1>A heading</h1>
    <a href="/">Link text</a>
  </body>
</html>
```

```
let heading = document.querySelector("h1");
```

```
heading.textContent = "New title!";
```

// After...

```
<html>
  <body>
    <h1>New title!</h1>
    <a href="/">Link text</a>
  </body>
</html>
```



Changing elements

// Before...

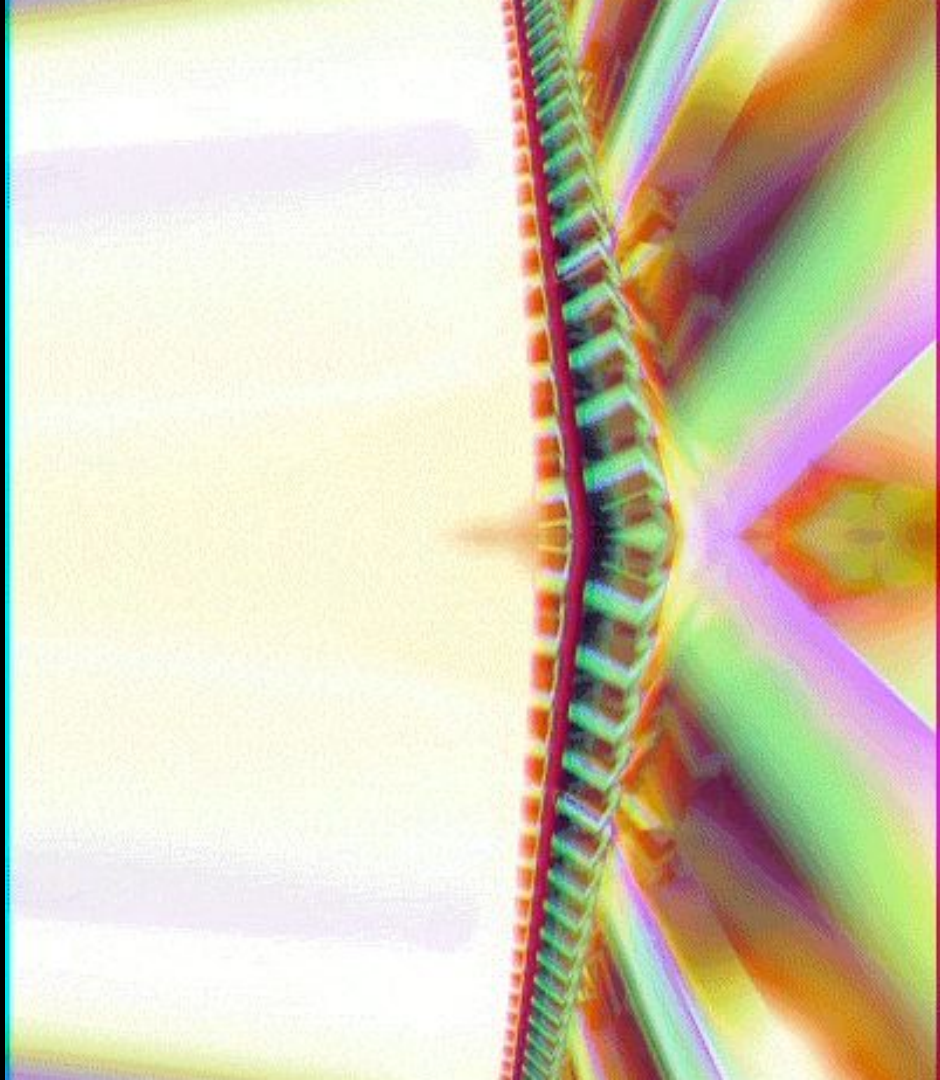
```
<html>
  <body>
    <h1>A heading</h1>
    <a href="/">Link text</a>
  </body>
</html>
```

```
let heading = document.querySelector("h1");
```

```
heading.innerHTML = "<span>Hello</span>";
```

// After...

```
<html>
  <body>
    <h1>
      <span>Hello</span>
    </h1>
    <a href="/">Link text</a>
  </body>
</html>
```



Changing styles

```
let heading = document.querySelector("h1");
```

```
// Cool...
```

```
heading.style.color = "hotpink";
```

```
heading.style.fontSize = "48px";
```

```
// Even better...
```

```
heading.classList.add("highlight");
```

```
// The HTML
```

```
<html>
```

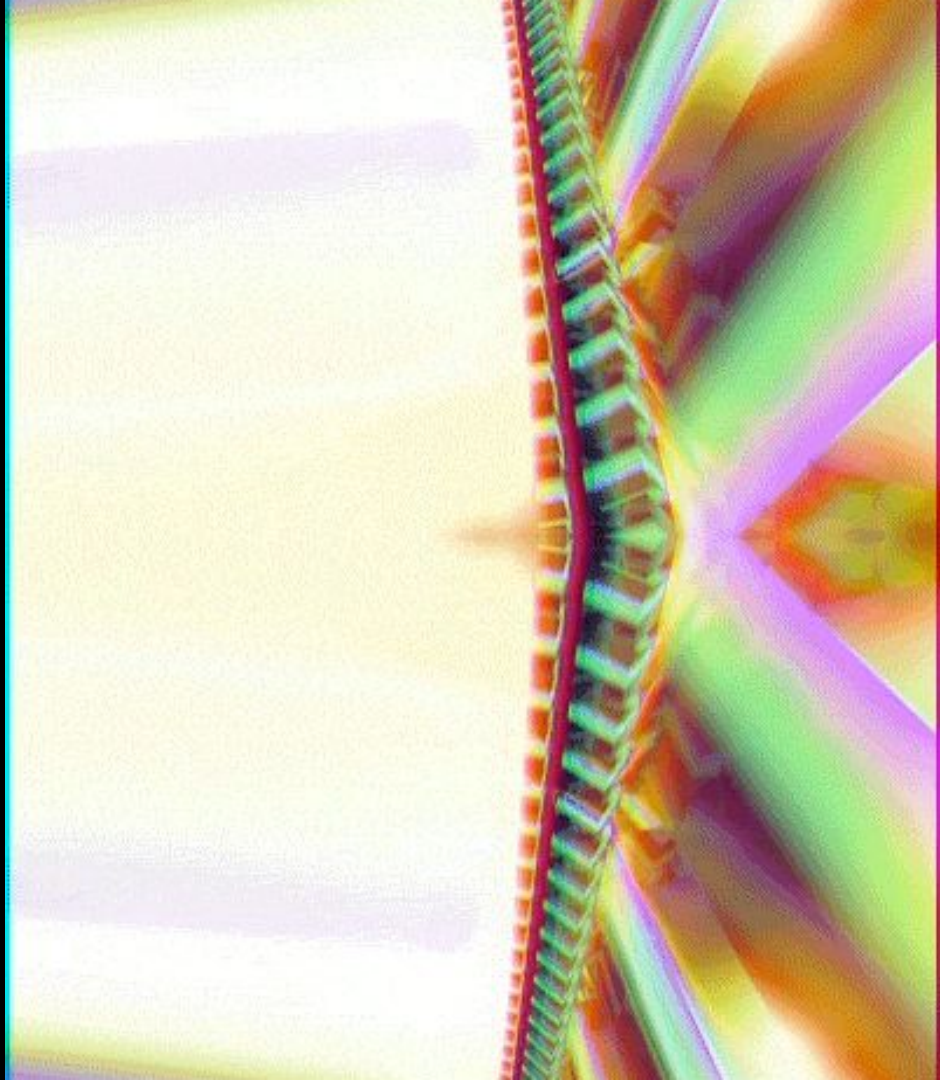
```
  <body>
```

```
    <h1 class="highlight">Title</h1>
```

```
    <a href="/">Link text</a>
```

```
  </body>
```

```
</html>
```



Changing attributes

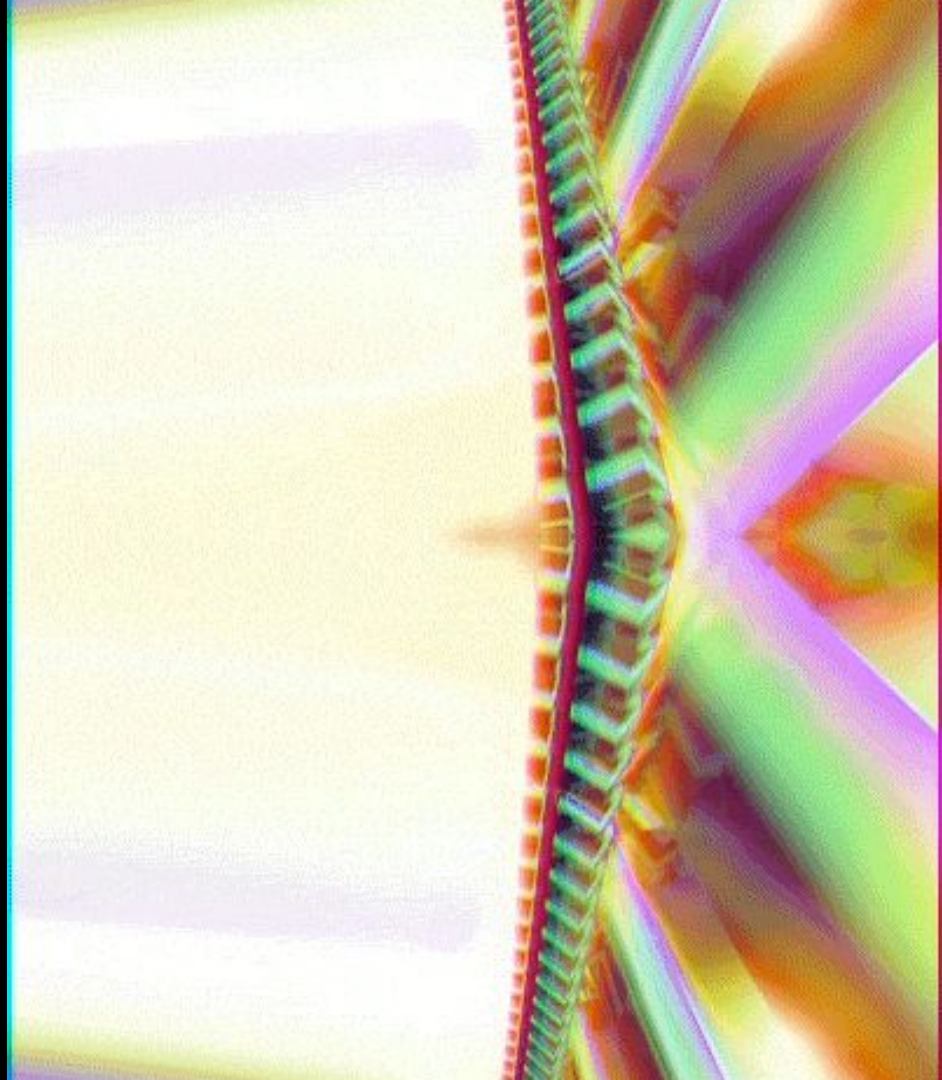
// Before...

```
<html>
  <body>
    <h1 class="highlight">Title</h1>
    <a href="/">Link text</a>
  </body>
</html>
```

```
let link = document.querySelector("a");
link.href = "bcit.ca";
link.textContent = "BCIT";
```

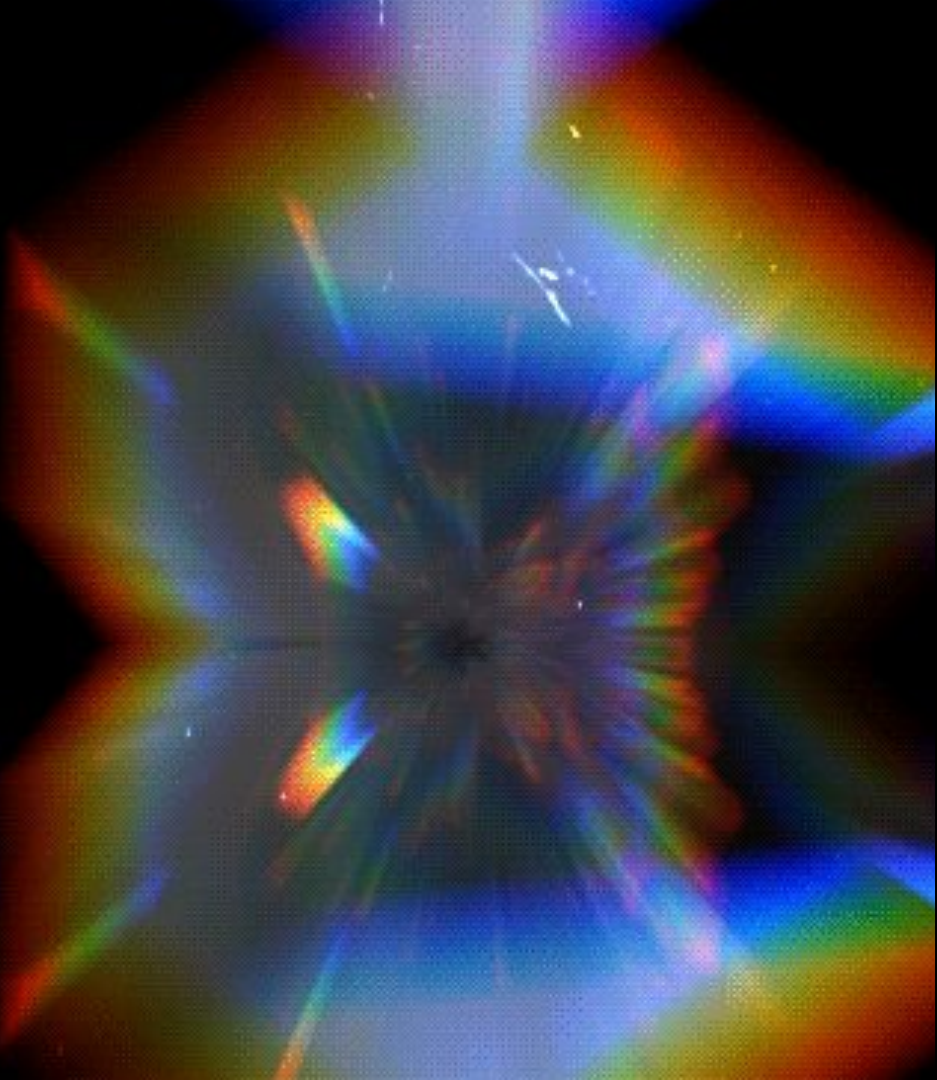
// After...

```
<html>
  <body>
    <h1 class="highlight">Title</h1>
    <a href="bcit.ca">BCIT</a>
  </body>
</html>
```



The DOM can also “listen” for events that trigger actions.

What is an event?



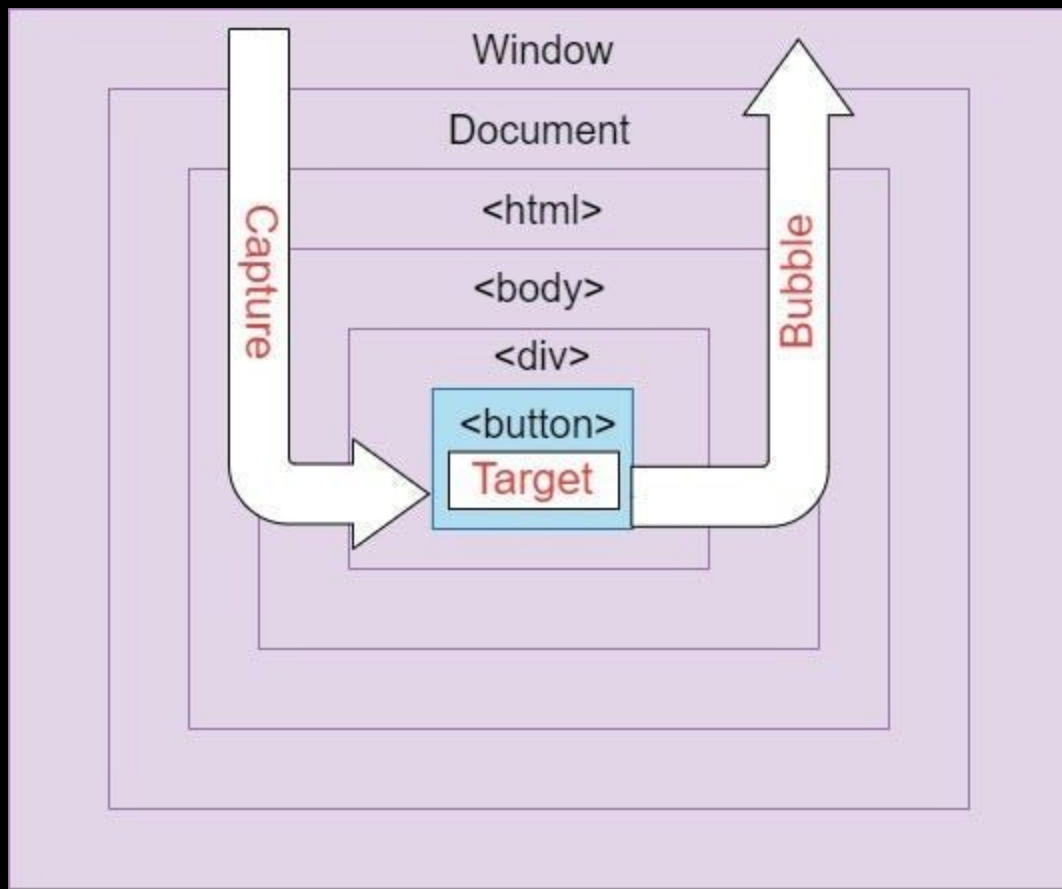
Events

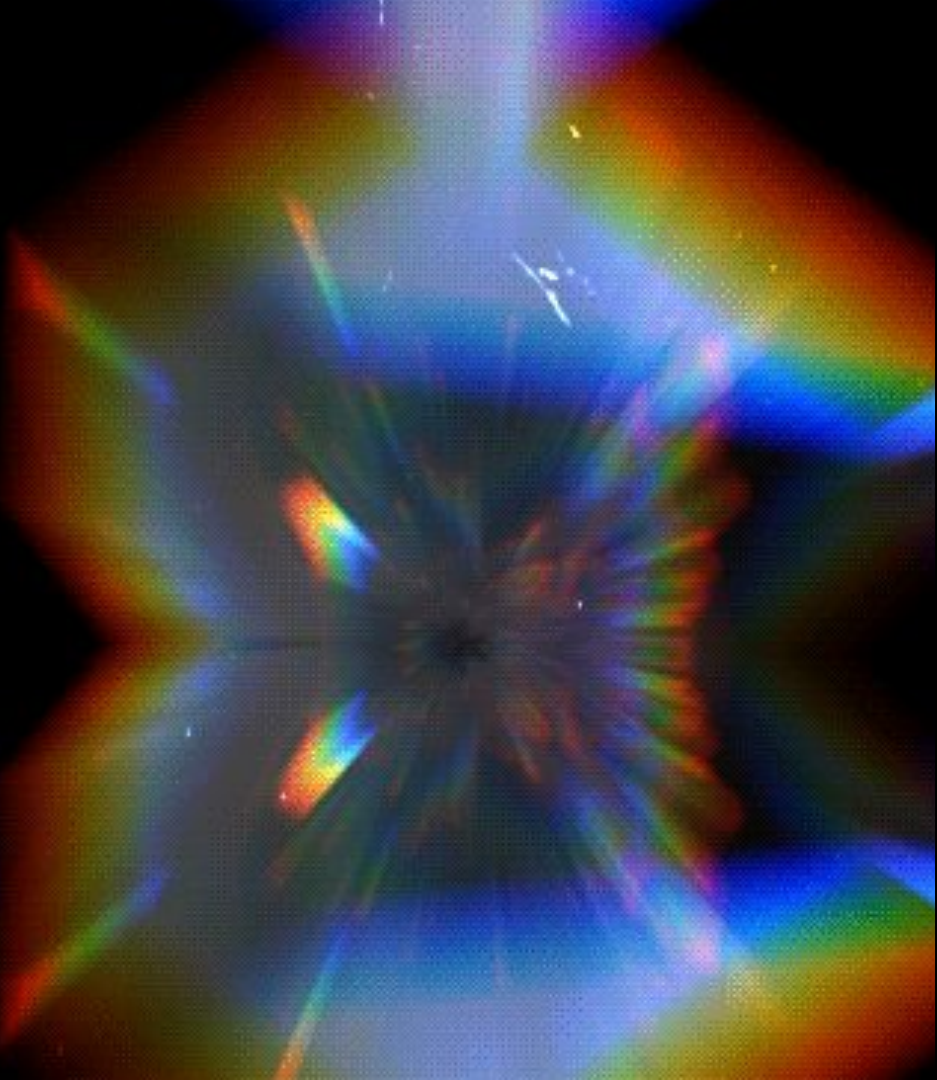
An event is something the user does:

- Click
- Type
- Scroll

Events connect user actions to JavaScript logic.

Be careful, because events can “bubble” up the DOM.





Events

```
<html>
  <body>
    <h1>A heading</h1>
    <a href="/">Link text</a>
  </body>
</html>
```

```
let link = document.querySelector("a");

link.addEventListener("click", () => {
  alert("Clicked!");
});
```